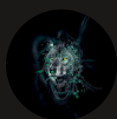




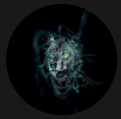
Testing VPC Connectivity



Kiprono Yegon

```
BWS [ALT+S]
}
.no-js-description {
  font-size: 1.25rem;
  color: #007286;
  line-height: 1.6;
  margin: 0;
  width: 80%;
}
.no-js-shield {
  width: 2.4rem;
  height: 2.4rem;
}
.no-js-logo-image {
  width: 2rem;
  height: 2rem;
}
@media (max-width: 768px) {
  .no-js-heading {
    font-size: 2.5rem;
  }
  .no-js-description {
    font-size: 1.125rem;
    width: 100%;
  }
  .no-js-container {
    padding: 0 1rem;
  }
}
</style>
<div class="no-js-overlay">
  <div class="no-js-logo">
    
  </div>
  <div class="no-js-container">
    
    <h1 class="no-js-heading">JavaScript is required, whaaaaaaat!</h1>
    <p class="no-js-description">
      NextWork requires JavaScript to function properly. Please enable
      JavaScript or disable any script-blocking tools like Brave Shields
      to use this site.
    </p>
  </div>
</div>
</noscript>
</body>
</html>
[ec2-user@ip-10-0-0-200 ~]$
```

i-05b4b86ed357c6e97 (NextWork Public Server)
PublicIPs: 35.91.160.1 PrivateIPs: 10.0.0.200



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a service from Amazon Web Services that allows you to create a private, isolated network within the cloud where you can securely run your applications and resources. It is useful because it gives you full control over your networking environment, including IP address ranges, subnets, routing, and security settings, enabling you to protect your systems, manage traffic flow, and design scalable architectures.

How I used Amazon VPC in this project

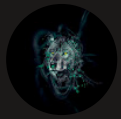
In today's project, I used Amazon VPC to test connectivity of a private and a public server and troubleshooting errors that arise

One thing I didn't expect in this project was...

One thing I didn't expect in this project was the in-depth learning of Network ACLs and security groups and the setup for private and public servers to communicate in a VPC

This project took me...

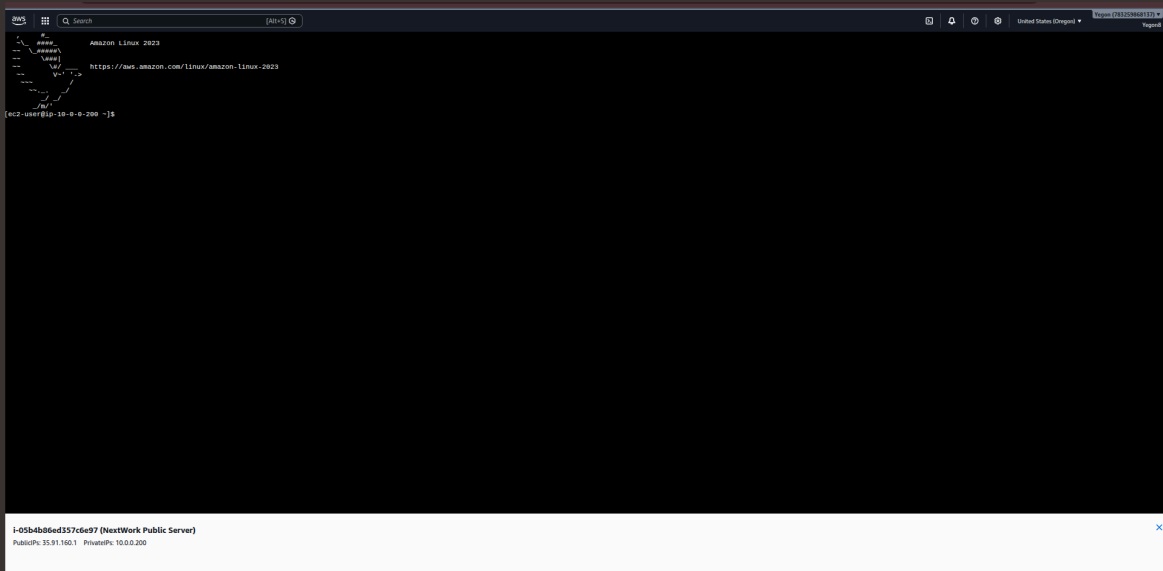
This project took me about 3 hours since I had previously deleted all the resources

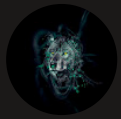


Connecting to an EC2 Instance

Connectivity is all about how well different parts of your network talk to each other and with external networks. It's essential because connectivity is how data flows smoothly across your network, powering everything from simple web hosting on the Internet to complex operations. Solid connectivity is the backbone of any system that relies on network interactions, making every communication and operation reliable and efficient.

My first connectivity test was whether I could connect to the Public Server by using the EC2 Instance Connect



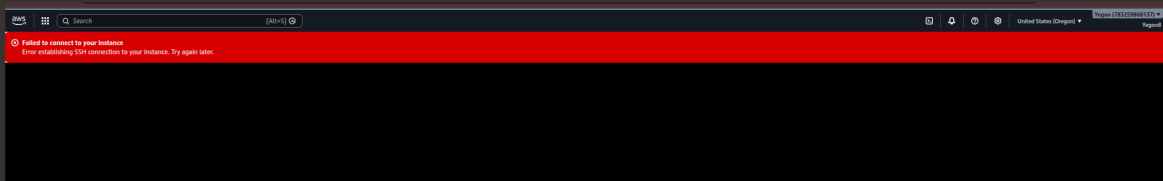


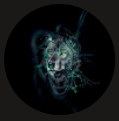
EC2 Instance Connect

I connected to my EC2 instance using EC2 Instance Connect, Instance Connect lets you securely connect to your EC2 instances directly using the AWS Management Console. You're still using SSH, but with all the key management handling it for you. This takes away a lot of the complexity of setting up SSH.

My first attempt at getting direct access to my public server resulted in an error, because the security group associated with NextWork Public Server lets in all inbound HTTP traffic, but this is not how we're trying to access our Public Server! We're trying to access NextWork Public Server using SSH through EC2 Instance Connect, which is a different traffic type and not the HTTP traffic we had added.

I fixed this error by adding a new rule in the inbound traffic. I added a nw rule that uses SSH traffic and chose "Anywhere IPv4 " as the source





Connectivity Between Servers

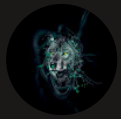
Ping is a common computer network tool used to check whether your computer can communicate with another computer or device on a network. I used ping to test the connectivity between the Public server and the Private server

The ping command I ran was "ping 10.0.1.78"

The first ping returned nothing, the screen was frozen. This meant there was no connectivity between the Public server and the Private server



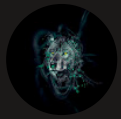
```
aws [ec2-user@ip-10-0-0-200 ~]$ ping 10.0.1.78
PING 10.0.1.78 (10.0.1.78) 56(84) bytes of data.
```



Troubleshooting Connectivity

I troubleshooted this by editing the network ACL to allow ICMP inbound from the Public server. Also updated the Private security group inbound rules to allow connection from the Public server

```
aws [Search] [Alt+5]
[ec2-user@ip-10.0.0.200 ~]$ ping 10.0.1.78
PING 10.0.1.78 (10.0.1.78) 56(84) bytes of data:
64 bytes from 10.0.1.78: icmp_seq=401 ttl=127 time=1.70 ms
64 bytes from 10.0.1.78: icmp_seq=402 ttl=127 time=0.315 ms
64 bytes from 10.0.1.78: icmp_seq=403 ttl=127 time=1.58 ms
64 bytes from 10.0.1.78: icmp_seq=404 ttl=127 time=0.985 ms
64 bytes from 10.0.1.78: icmp_seq=405 ttl=127 time=0.337 ms
64 bytes from 10.0.1.78: icmp_seq=406 ttl=127 time=1.89 ms
64 bytes from 10.0.1.78: icmp_seq=407 ttl=127 time=0.314 ms
64 bytes from 10.0.1.78: icmp_seq=408 ttl=127 time=1.40 ms
64 bytes from 10.0.1.78: icmp_seq=409 ttl=127 time=0.301 ms
64 bytes from 10.0.1.78: icmp_seq=500 ttl=127 time=2.11 ms
64 bytes from 10.0.1.78: icmp_seq=501 ttl=127 time=0.513 ms
64 bytes from 10.0.1.78: icmp_seq=502 ttl=127 time=1.45 ms
64 bytes from 10.0.1.78: icmp_seq=503 ttl=127 time=0.608 ms
64 bytes from 10.0.1.78: icmp_seq=504 ttl=127 time=1.33 ms
64 bytes from 10.0.1.78: icmp_seq=505 ttl=127 time=0.550 ms
64 bytes from 10.0.1.78: icmp_seq=506 ttl=127 time=1.73 ms
64 bytes from 10.0.1.78: icmp_seq=507 ttl=127 time=0.744 ms
64 bytes from 10.0.1.78: icmp_seq=508 ttl=127 time=3.57 ms
64 bytes from 10.0.1.78: icmp_seq=509 ttl=127 time=1.31 ms
64 bytes from 10.0.1.78: icmp_seq=510 ttl=127 time=0.653 ms
64 bytes from 10.0.1.78: icmp_seq=511 ttl=127 time=1.68 ms
64 bytes from 10.0.1.78: icmp_seq=512 ttl=127 time=0.646 ms
64 bytes from 10.0.1.78: icmp_seq=513 ttl=127 time=1.33 ms
64 bytes from 10.0.1.78: icmp_seq=514 ttl=127 time=0.805 ms
64 bytes from 10.0.1.78: icmp_seq=515 ttl=127 time=1.49 ms
64 bytes from 10.0.1.78: icmp_seq=516 ttl=127 time=0.666 ms
64 bytes from 10.0.1.78: icmp_seq=517 ttl=127 time=1.86 ms
64 bytes from 10.0.1.78: icmp_seq=518 ttl=127 time=0.460 ms
64 bytes from 10.0.1.78: icmp_seq=519 ttl=127 time=1.71 ms
64 bytes from 10.0.1.78: icmp_seq=520 ttl=127 time=0.318 ms
64 bytes from 10.0.1.78: icmp_seq=521 ttl=127 time=2.23 ms
64 bytes from 10.0.1.78: icmp_seq=522 ttl=127 time=0.298 ms
64 bytes from 10.0.1.78: icmp_seq=523 ttl=127 time=2.04 ms
64 bytes from 10.0.1.78: icmp_seq=524 ttl=127 time=0.879 ms
64 bytes from 10.0.1.78: icmp_seq=525 ttl=127 time=0.347 ms
64 bytes from 10.0.1.78: icmp_seq=526 ttl=127 time=1.44 ms
64 bytes from 10.0.1.78: icmp_seq=527 ttl=127 time=0.337 ms
```



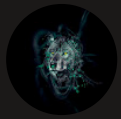
Connectivity to the Internet

Curl is a tool to test connectivity in a network, curl is used to transfer data to or from a server. That means on top of checking connectivity, you can use curl to grab data from, or upload data into other servers on the internet

I used curl to test the connectivity between the Public server and example.com, curl has fetched the complete HTML content of web app, which is why you now see a large amount of HTML data.

Ping vs Curl

Ping and curl are different because curl tests connectivity in a network whereas ping checks if one computer can contact another and how long messages take to travel back and forth



Connectivity to the Internet

I ran the curl command `curl learn.nextwork.org` the command sends an HTTP request to the server that hosts the website. This request tells the server that you want to retrieve the HTML content of the website. The website's server then processes this request and sends back the data as a response.

```
aws
Q Search [Alt+S]
}
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  font-size: 1.25rem;
  color: #007280;
  line-height: 1.6;
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PublicIPs: 35.91.160.1 PrivateIPs: 10.0.0.200



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